AFRICAST

TIDY TIME SERIES & FORECASTING IN R

Basic training and test accuracy



Session 8 of 10

SECTION 1 OF 4

Evaluating forecast accuracy

1 **Evaluating forecast accuracy**
2 Evaluating point forecast accuracy
3 Evaluating distributional forecast accuracy
4 Forecast ensembles



Evaluate forecast accuracy

- In order to evaluate the performance of a forecasting model, we compute its forecast accuracy.
- Forecast accuracy is compared by measuring errors based on the test set.
- Ideally, compare the benefit of improved accuracy with the cost of obtaining it.

Evaluate forecast accuracy: decision impact

Best forecast = best decisions

Choose the model whose forecasts lead to better decisions, such as:

- lower staffing costs
- lower emissions
- higher service levels
- fewer stock-outs
- less excess inventory
- faster response
- more stable plans

- Decision impact is often hard to measure.
- Fallback: choose the most accurate forecast.

Which error costs more?

In your work, which forecast error is more costly?

Forecasting too high (waste, idle staff, excess stock)	0	0%
Forecasting too low (stock-outs, unmet need, overloaded services)	0	0%
About the same	0	0%
It depends on the situation	0	0%



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0 responses

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A perfect fit

A model fits the training data perfectly. Will it forecast well?

Yes	0	0%
Not necessarily	0	0%
No	0	0%



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africast-accuracy

0 responses

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In-sample (training) vs. out-of-sample (test)

- Residuals (in-sample errors) are not a reliable indication of forecast accuracy
- A model which fits the training data well will not necessarily forecast well
- A perfect fit can always be obtained by using a model with enough parameters
- Over-fitting a model to data is just as bad as failing to identify a systematic pattern in the data

How long a test set?

You need forecasts 12 months ahead. What is the shortest sensible test set?

3 months	**0**	0%
12 months	**0**	0%
36 months	**0**	0%

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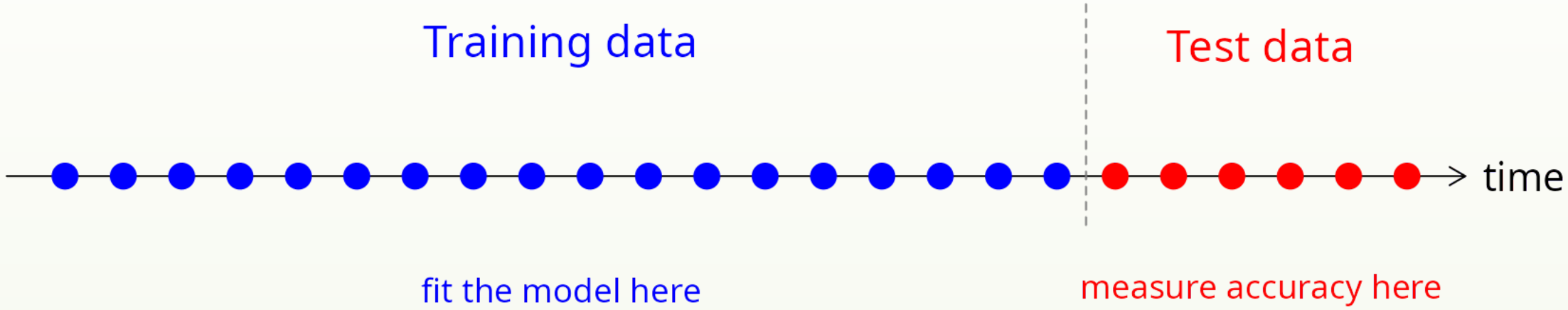
0 responses

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Forecast accuracy evaluation using test sets

- We mimic the real life situation
- We pretend we don’t know some part of data (new data)
- It must not be used for *any* aspect of model training
- Forecast accuracy is computed only based on the test set
- The test set should ideally be at least as long as the maximum forecast horizon

Training and test sets



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Evaluating point forecast accuracy

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Forecast errors

Forecast "error": the difference between an observed value and its forecast.

$$e_{T+h} = y_{T+h} - \hat{y}_{T+h|T},$$

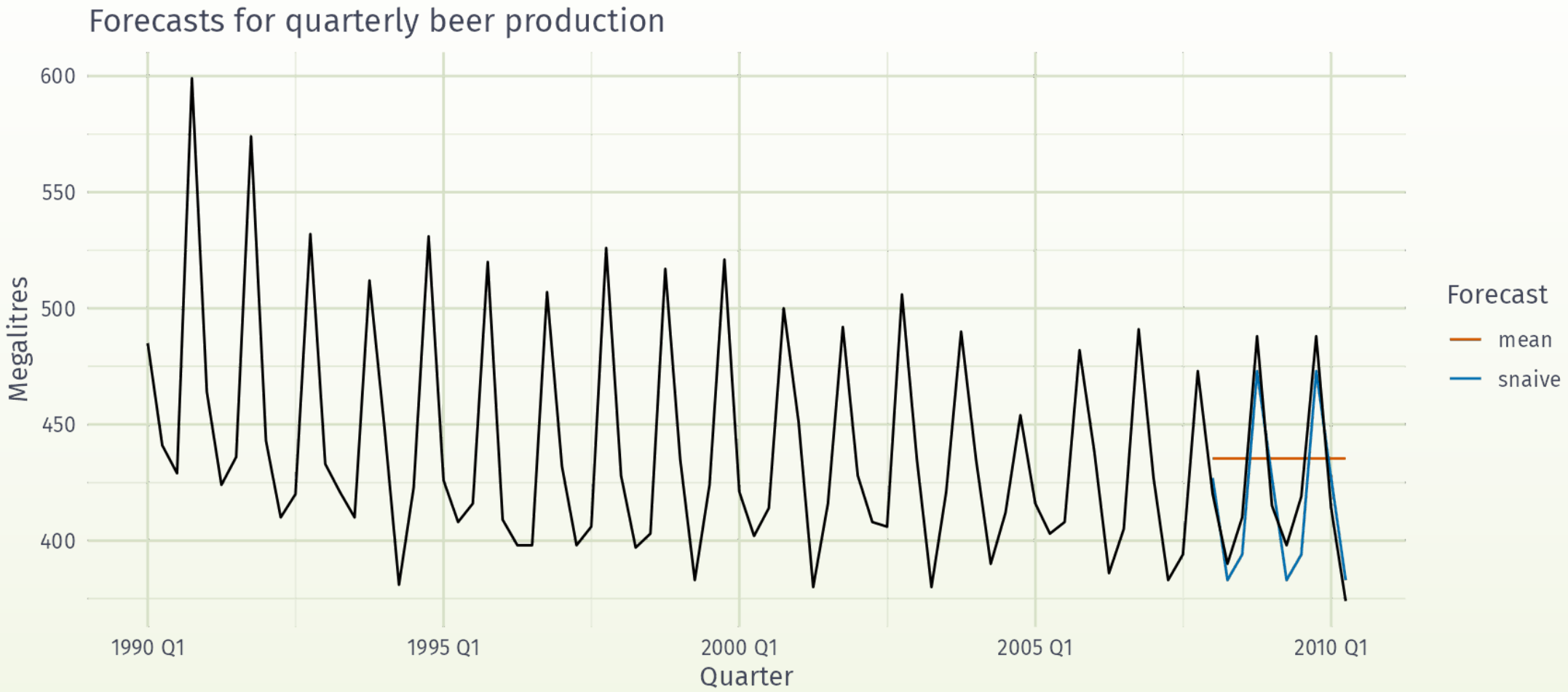
where

- the training data is $\{y_1, \ldots, y_T\}$
- y_{T+h} is the hth test observation, $h = 1, \ldots, H$
- $\hat{y}_{T+h|T}$ is its forecast based on data up to time T

Forecasting beer production

```
beer_fit <- aus_production |>
  filter(between(year(Quarter), 1992, 2007)) |>
  model(
    snaive = SNAIVE(Beer),
    mean = MEAN(Beer)
  )
beer_fit |>
  forecast(h = 10) |>
  autoplot(filter(aus_production, year(Quarter) >= 1990), level = NULL) +
  labs(
    title = "Forecasts for quarterly beer production",
    x = "Quarter", y = "Megalitres"
  ) +
  guides(colour = guide_legend(title = "Forecast"))
```

Forecasting beer production



Scale-dependent and percentage errors

$$\begin{aligned}
\text{MAE} &= \text{mean}(|e_{T+h}|) \\
\text{MSE} &= \text{mean}(e_{T+h}^2) \\
\text{RMSE} &= \sqrt{\text{mean}(e_{T+h}^2)} \\
\text{MAPE} &= 100\,\text{mean}(|e_{T+h}|/|y_{T+h}|)
\end{aligned}$$

- MAE, MSE, RMSE are all scale dependent.
- MAPE is scale independent but is only sensible if $y_t \gg 0$ for all t, and y has a natural zero.

Comparing across districts

You forecast clinic visits in 50 districts of very different sizes. Which measure lets you compare your method’s accuracy across all of them?

RMSE **0** 0%

MAE **0** 0%

MASE **0** 0%



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0 responses

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Scaled errors: MASE

Mean Absolute Scaled Error

$$\text{MASE} = \text{mean}(|e_{T+h}|/Q)$$

- For non-seasonal series, scale uses naïve forecasts:

$$Q = \frac{1}{T-1}\sum_{t=2}^{T}|y_t - y_{t-1}|$$

- For seasonal series, scale uses seasonal naïve forecasts:

$$Q = \frac{1}{T-m}\sum_{t=m+1}^{T}|y_t - y_{t-m}|$$

where m is the seasonal frequency, and Q is computed on the training data

Proposed by Hyndman and Koehler (IJF, 2006).

Slow-selling items

Which measure suits daily sales of a slow-selling item with many zero days?

MAPE	0	0%
MASE	0	0%
Either	0	0%



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0 responses

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Scaled errors: RMSSE

Root Mean Squared Scaled Error

$$\text{RMSSE} = \sqrt{\text{mean}(e_{T+h}^2/Q)}$$

- For non-seasonal series, scale uses naïve forecasts:

$$Q = \frac{1}{T-1}\sum_{t=2}^{T}(y_t - y_{t-1})^2$$

- For seasonal series, scale uses seasonal naïve forecasts:

$$Q = \frac{1}{T-m}\sum_{t=m+1}^{T}(y_t - y_{t-m})^2$$

where m is the seasonal frequency, and Q is computed on the training data

Used in the M5 forecasting competition (Makridakis et al., IJF, 2022).

Test-set accuracy

```
beer_fc <- forecast(beer_fit, h = 10)
beer_fc |>
  accuracy(
    aus_production,
    measures = list(RMSE = RMSE, MAE = MAE, MAPE = MAPE, MASE = MASE, RMSSE = RMSSE)
  )
```

```
# A tibble: 2 × 7
  .model .type  RMSE   MAE  MAPE  MASE RMSSE
  <chr>  <chr> <dbl> <dbl> <dbl> <dbl> <dbl>
1 mean   Test   38.4  34.8  8.28 2.20  1.96
2 snaive Test   14.3  13.4  3.17 0.847 0.729
```

- Seasonal naïve beats the mean on every measure.
- Its MASE and RMSSE are below 1: smaller errors than in-sample seasonal naïve.

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Evaluating distributional forecast accuracy

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Prediction interval accuracy using Winkler score

Winkler (1972) proposed a scoring method to enable comparisons between prediction intervals:

- it takes account of both coverage and width of the intervals.

Winkler score

$$W(l_t, u_t, y_t) = \begin{cases} u_t - l_t & \text{if } l_t \leq y_t \leq u_t \\ (u_t - l_t) + \frac{2}{\alpha}(l_t - y_t) & \text{if } y_t < l_t \\ (u_t - l_t) + \frac{2}{\alpha}(y_t - u_t) & \text{if } y_t > u_t \end{cases}$$

where $[l_t, u_t]$ is a $100(1-\alpha)\%$ prediction interval.

- Lower is better.
- Misses are penalised by $2/\alpha$ times the distance outside the interval.

Prediction interval accuracy

```
beer_fc |>
  accuracy(aus_production, measures = list(winkler = winkler_score), level = 95)
```

```
# A tibble: 2 × 3
  .model .type winkler
  <chr>  <chr>   <dbl>
1 mean   Test    174.
2 snaive Test     86.3
```

- Winkler scores for 95% prediction intervals ($\alpha = 0.05$).
- Both cover every test observation, but seasonal naïve intervals are half as wide.

Quantile score

Quantile score

$$Q_{p,t} = \begin{cases} 2(1-p)\big(f_{p,t} - y_t\big), & \text{if } y_t < f_{p,t} \\ 2p\big(y_t - f_{p,t}\big), & \text{if } y_t \geq f_{p,t} \end{cases}$$

where $f_{p,t}$ is the p-quantile of the forecast distribution for time t.

- Lower is better.

Quantile score

```
beer_fc |>
  accuracy(aus_production, list(qs = quantile_score), probs = 0.9)
```

```
# A tibble: 2 × 3
  .model .type    qs
  <chr>  <chr> <dbl>
1 mean   Test  14.1
2 snaive Test   4.60
```

- Quantile score of the 90th percentile forecasts ($p = 0.9$).

Continuous Ranked Probability Score (CRPS)

$$\mathrm{CRPS} = \mathrm{mean}(p_t),$$

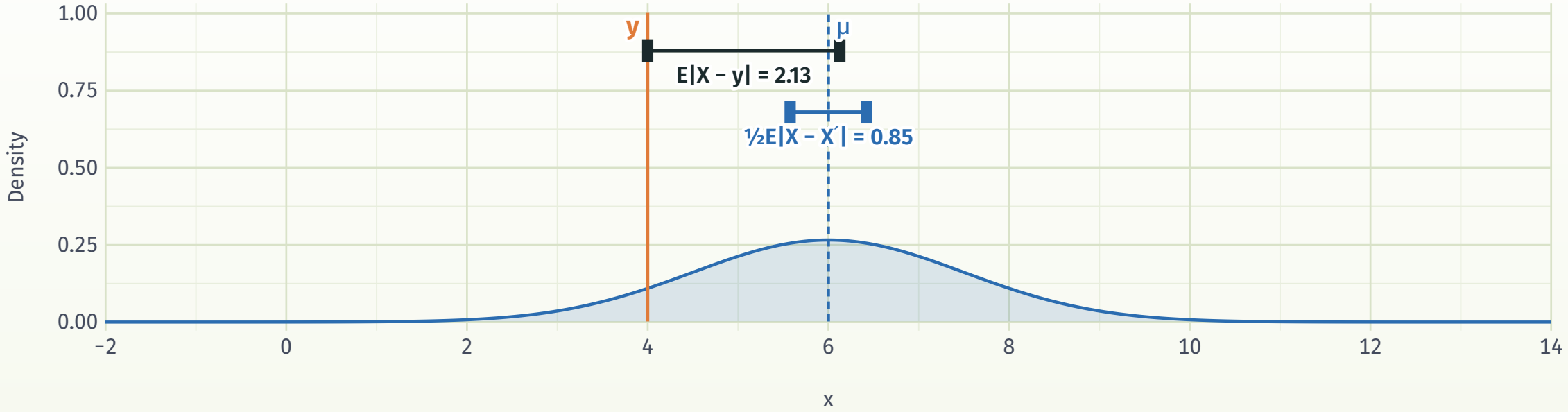
where

$$p_t = \int_{-\infty}^{\infty} \left(G_t(x) - F_t(x)\right)^2 dx$$

- $G_t(x)$ is the forecast distribution function for time t
- $F_t(x)$ is the observation's distribution function: a step from 0 to 1 at y_t
- Lower is better; for a point forecast it reduces to the MAE.

CRPS: expected error less spread

CRPS = 2.13 − 0.85 = **1.28**



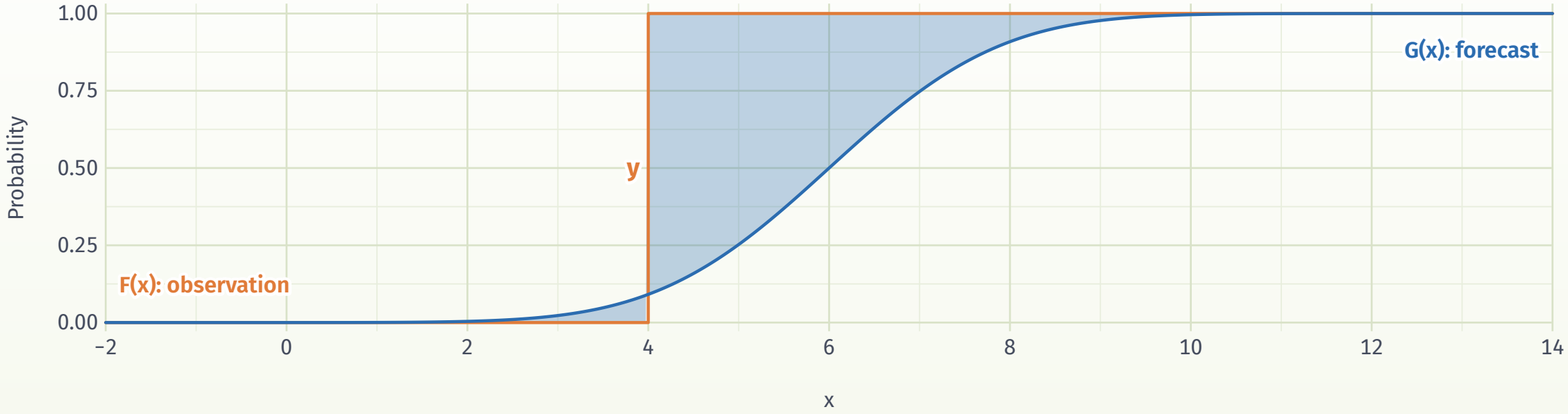
Forecast mean (μ) 6.0 (0, 2, 4, 6, 8, 10)

Forecast sd (σ) 1.5 (0.5, 1, 1.5, 2, 2.5, 3)

- Equivalently, CRPS $= \mathrm{E}|X - y| - \frac{1}{2}\mathrm{E}|X - X'|$, where X, X' are independent draws from G.
- Expected distance from y, less half the forecast's spread.

CRPS: the gap between forecast and outcome

CRPS = **1.28**



Forecast mean (μ) 6.0

0 2 4 6 8 10

Forecast sd (σ) 1.5

0.5 1 1.5 2 2.5 3

- CRPS adds up the squared gap (shaded) between the forecast G and the outcome F.
- Mean away from y, or spread too wide: larger. Lower is better.

Percentile score and CRPS

```
beer_fc |>
  accuracy(aus_production, measures = distribution_accuracy_measures)
```

```
# A tibble: 2 × 4
  .model .type percentile  CRPS
  <chr>  <chr>      <dbl> <dbl>
1 mean   Test       22.7  22.4
2 snaive Test        8.80  8.71
```

- Percentile score: quantile score averaged over $p = 0.01, 0.02, \ldots, 0.99$.
- Seasonal naïve has the better distributional forecasts on both measures.

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Forecast ensembles

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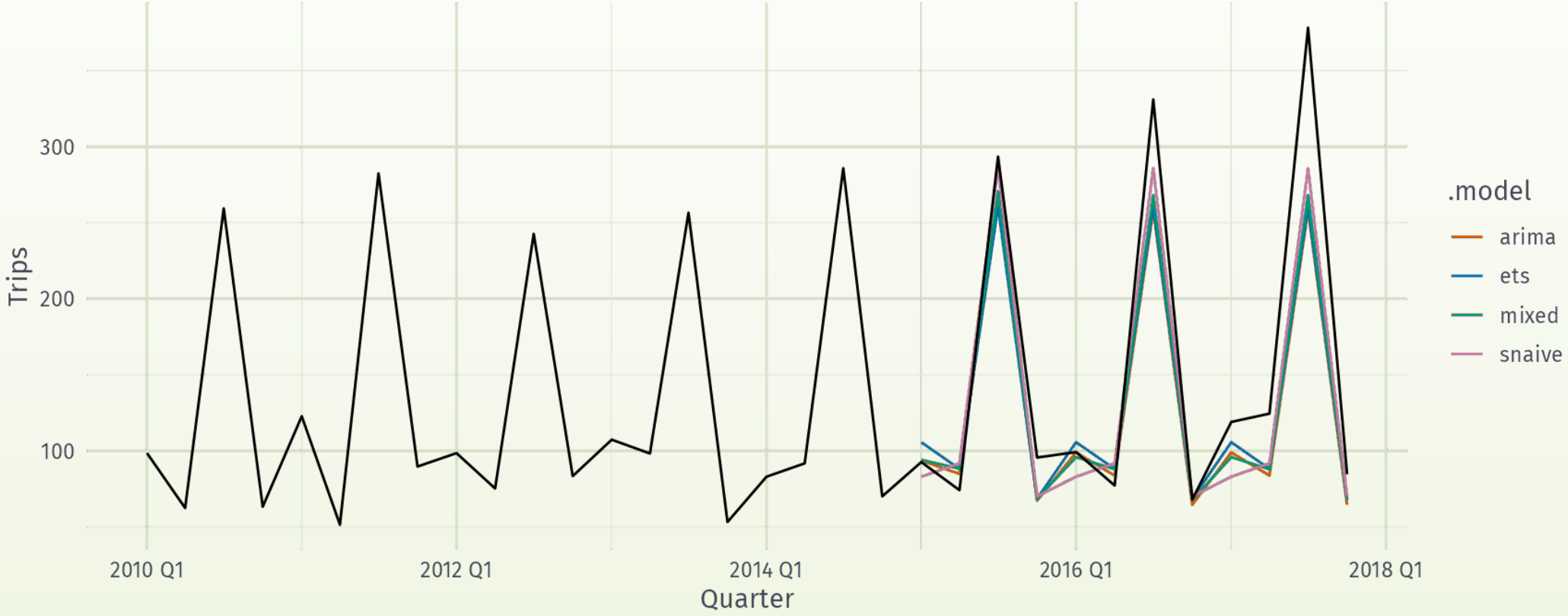
Forecast ensembles

```
train <- tourism |>
  filter(year(Quarter) <= 2014)
fit_tourism <- train |>
  model(
    ets = ETS(Trips),
    arima = ARIMA(Trips),
    snaive = SNAIVE(Trips)
  ) |>
  mutate(mixed = (ets + arima + snaive) / 3)
```

- Fits three models to each of the 304 tourism series (this takes a while).
- Ensemble forecast `mixed` is a simple average of the three fitted models.
- `forecast()` will produce distributional forecasts taking into account the correlations between the forecast errors of the component models.

Forecast ensembles

```
fc <- fit_tourism |> forecast(h = "3 years")
fc |>
  filter(Region == "Snowy Mountains", Purpose == "Holiday") |>
  autoplot(tourism |> filter(year(Quarter) >= 2010), level = NULL)
```



Forecast ensembles

Accuracy on the 2015–2017 test set, averaged across all series.

```
fc |>
  accuracy(tourism) |>
  group_by(.model) |>
  summarise(RMSE = mean(RMSE), MASE = mean(MASE)) |>
  arrange(RMSE)
```

```
# A tibble: 4 × 3
  .model  RMSE  MASE
  <chr>  <dbl> <dbl>
1 mixed   19.8 0.997
2 ets     20.2 1.00
3 snaive  21.5 1.17
4 arima   21.9 1.06
```

Averaging lets errors from different models partially cancel, and reduces the risk of picking the wrong model.